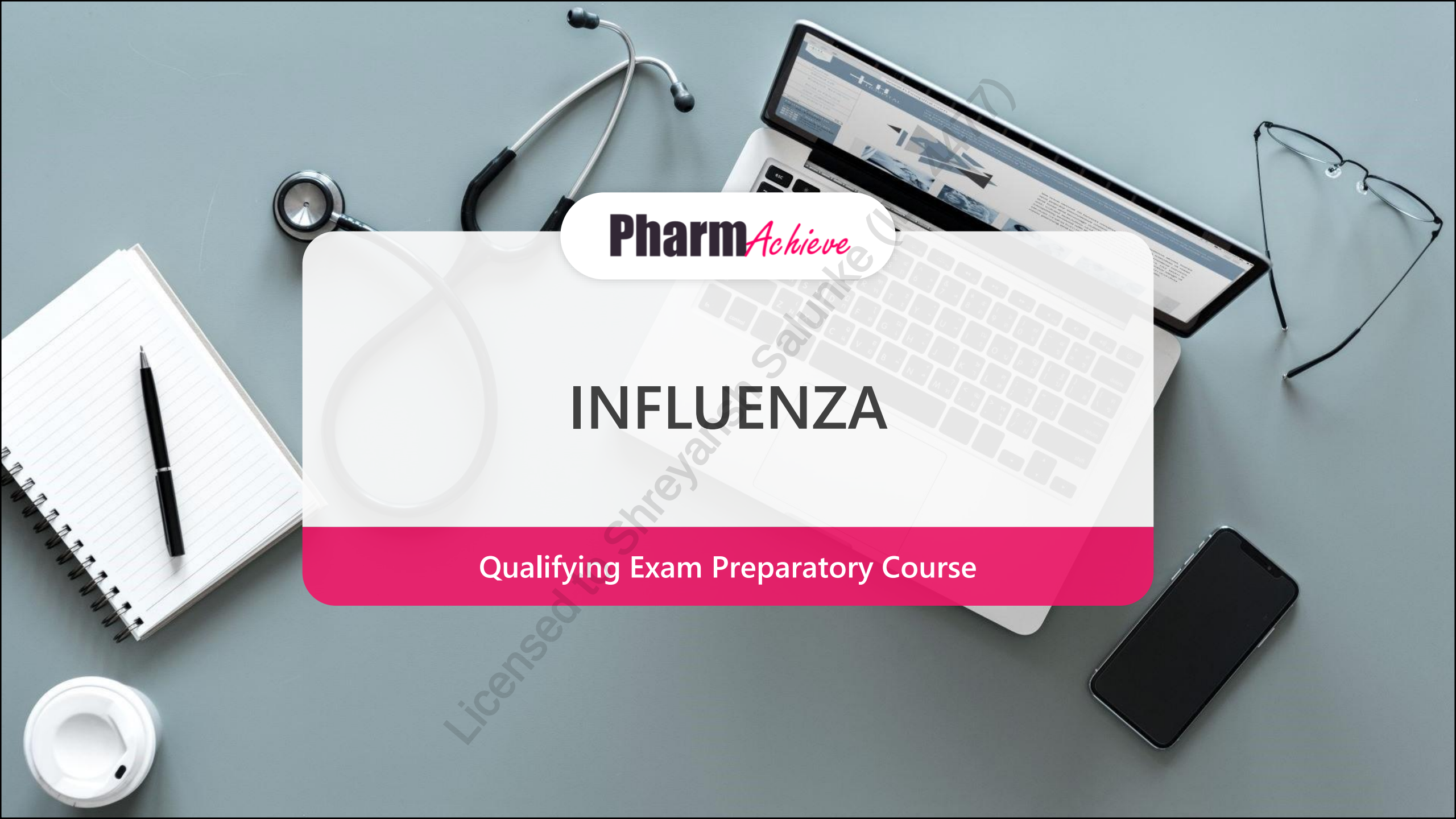




Pharm*Achieve*

INFLUENZA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ASA	Aspirin	ILI	Influenza-Like Illness
CrCl	Creatinine Clearance	LAIV	Live Attenuated Influenza Vaccine
COPD	Chronic Obstructive Pulmonary Disease	LAIV4	Quadrivalent live attenuated influenza vaccine
CI	Contraindicated	mQIV	Mammalian cell culture-based Quadrivalent Inactivated Influenza Vaccine
GBS	Guillain-Barré syndrome	NACI	National Advisory Committee on Immunization
HF	heart failure	PCR	Polymerase Chain Reaction
IDSA	Infections Diseases Society of America	PEP	Post-exposure prophylaxis
IIV3-Adj	Adjuvanted trivalent inactivated influenza vaccine	QIV	Quadrivalent Inactivated Influenza Vaccine
IIV3-SD	Standard-dose trivalent inactivated influenza vaccine	RAT	Rapid Antigen Test
IIV4-cc	Quadrivalent mammalian cell culture-based inactivated influenza vaccine	RIV4	Quadrivalent recombinant influenza vaccine
IIV4-HD	High-dose quadrivalent inactivated influenza vaccine	SJS	Stevens-Johnson Syndrome
IIV4-SD	Standard-dose quadrivalent inactivated influenza vaccine	TIV	Trivalent Inactivated Influenza Vaccine

DEFINITIONS

Influenza

- Acute upper respiratory tract infection caused primarily by influenza A and/or influenza B viruses during seasonal flu outbreaks and less commonly by influenza C

Influenza Season

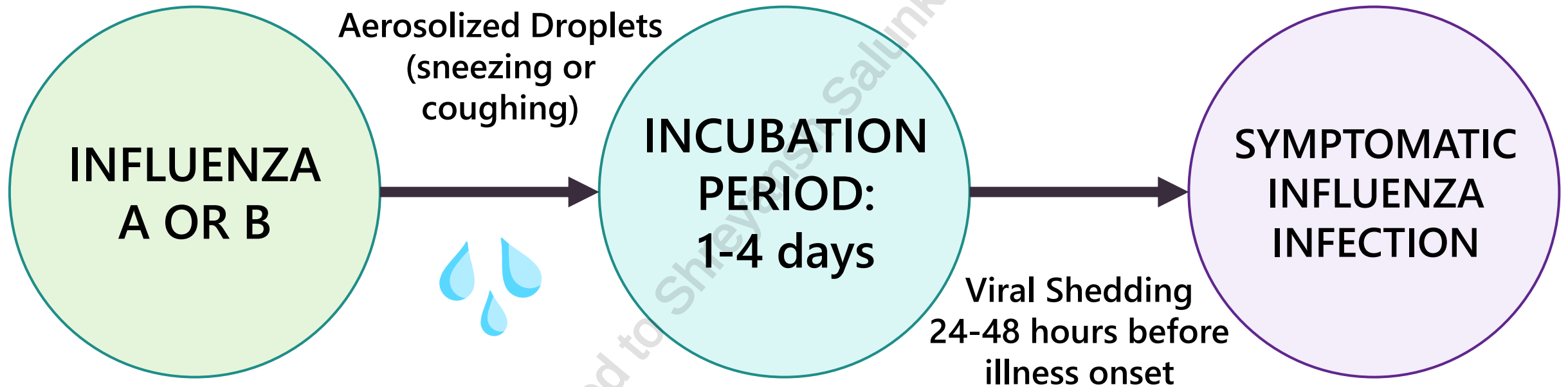
- Surveillance period when influenza activity typically occurs
- In North America, the influenza season typically starts in November through February to March

Influenza-Like Illness (ILI)

- Refers to any acute illness that has the symptoms of influenza
- Similar symptoms can be caused by adenovirus, coronavirus, parainfluenza or respiratory syncytial virus

PATHOPHYSIOLOGY

FYI



CLINICAL PRESENTATION

Diagnosis is often reliant on clinical presentation.
Influenza begins with an acute onset of:

- Fever
- Headache
- Myalgias
- Malaise
- Cough
- Sore throat
- Nasal congestion

- The symptoms of influenza last about 4-7 days
- Gastrointestinal symptoms can occur in children <5 years of age
- Fever may not be a prominent symptom in children <5 years or adults ≥65 years of age

PATIENT ASSESSMENT: HISTORY

Healthy adolescents and adults

- When influenza is circulating in the community OR if the patient has returned from a travel area where influenza is circulating
- **Abrupt onset of fever $\geq 38^{\circ}\text{C}$**
- Symptoms: fever, headache, muscle aches, fatigue, cough and sore throat may be present

Young children

- May present with a vague febrile illness or a respiratory condition resembling croup, bronchitis, and sometimes bronchiolitis
- GI symptoms (e.g. nausea, vomiting and diarrhea) can also occur in children <5 years

Older adults

- Fever may not be prominent
- If influenza is circulating in the community OR if the individual has returned from a travel area where influenza is circulating, **suspect that the patient has influenza IF the patient presents with ≥ 2 of the following** sore throat, arthralgia, myalgia, cough, or prostration; this is true especially in elderly living in long-term care facilities

PATIENT ASSESSMENT: RED FLAGS



- Difficulty breathing/shortness of breath/rapid breathing
- Bluish skin colour or change in skin colour
- Confusion, listlessness, altered consciousness
- Dehydration
- Not waking up or not interacting
- Being so irritable that child does not want to be held
- Flu-like symptoms improve but then return with fever and a worse cough
- Fever with a rash
- Seizures

PATIENT ASSESSMENT: INFLUENZA CONFIRMATION

- Influenza testing is not recommended for patients who are not at high risk for influenza complications
- IDSA recommends testing for influenza A & B in the following **outpatients if the results will influence clinical management of the infection** during flu season:
 - A. High-risk patients (e.g. immunocompromised, children <5, adults ≥65, chronic health conditions) that are presenting with ILI, nonspecific respiratory illnesses or pneumonia
 - B. Patients presenting with acute respiratory symptoms who have either an exacerbation of medical conditions (e.g. HF, asthma, COPD) or known influenza complications (e.g. pneumonia)
- Specimens from the upper respiratory tract (e.g. nasal or nasopharyngeal swabs) should be collected as soon as possible, **ideally within 4 days of symptom onset**

GOALS OF THERAPY

- ✓ Prevent the onset of influenza, especially in patients at risk for complications
- ✓ Alleviate influenza symptoms
- ✓ Prevent the development of influenza complications, influenza-related hospitalizations and deaths
- ✓ Educate patients on the proper use of non-pharmacological therapy and antiviral medications
- ✓ Reduce the inappropriate use of antibiotics in viral respiratory infections

NON-PHARMACOLOGICAL & OVER-THE-COUNTER MEASURES

Reduce transmission	Symptomatic improvement
• Flu vaccine • Use of masks • Cough/sneeze into a tissue or a sleeve • Hand hygiene (soap + water or $\geq 60\%$ ethanol sanitizer) • Staying home/ social distancing	• Acetaminophen • Ibuprofen • Hydration • Rest • Saline nasal sprays • Cough and cold products (e.g. dextromethorphan, lozenges)

PHARMACOLOGIC THERAPY: ANTIVIRAL AGENTS

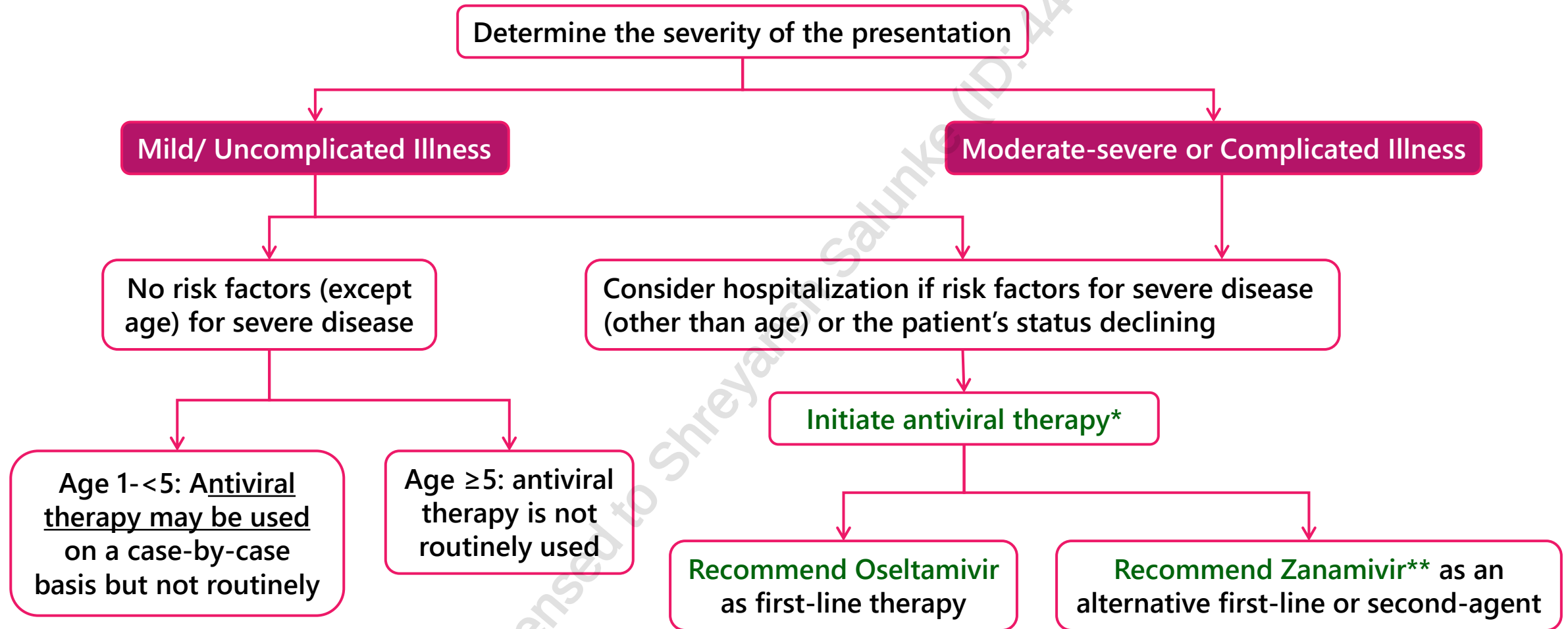
- Antiviral agents are not routinely recommended for treating influenza in healthy children and adults but can be considered on a case-by-case basis
- **Antiviral agents should be initiated** as soon as possible, ideally **within 48 hours of illness onset**, for the following patients:
 1. Age 1-64 with chronic disease or immunocompromised or high risk
 2. Age ≥ 65 with confirmed influenza
 3. Patients of any age requiring hospitalization due to the progression of the infection
- **Antiviral therapy** should be **initiated even if 48 hours have passed** in the following patients:
 1. Influenza virus results in the patient's hospitalization
 2. The patient is considered high risk for severe disease
 3. Despite being healthy, the disease state has progressed into a severe or complicated state

INDIVIDUALS WHO MAY BENEFIT FROM ANTIVIRAL AGENTS

Patients with the following conditions are at a **Higher Risk of Influenza Complications:**

- Chronic pulmonary conditions (e.g. asthma, COPD)
- Cardiovascular conditions (e.g. heart failure, excluding hypertension alone)
- Metabolic conditions (e.g. diabetes, obesity)
- Renal and/or hepatic insufficiency
- Neurological disorders (e.g. epilepsy, stroke, spinal cord injury)
- Immunocompromised status or immunosuppressant medications (e.g. cancer or HIV medications)
- Pregnancy or postpartum
- Long-term care residents
- Age ≤ 18 taking ASA therapy due to increased risk of experiencing Reyes syndrome after influenza infection
- Age < 5 and ≥ 65 with confirmed influenza

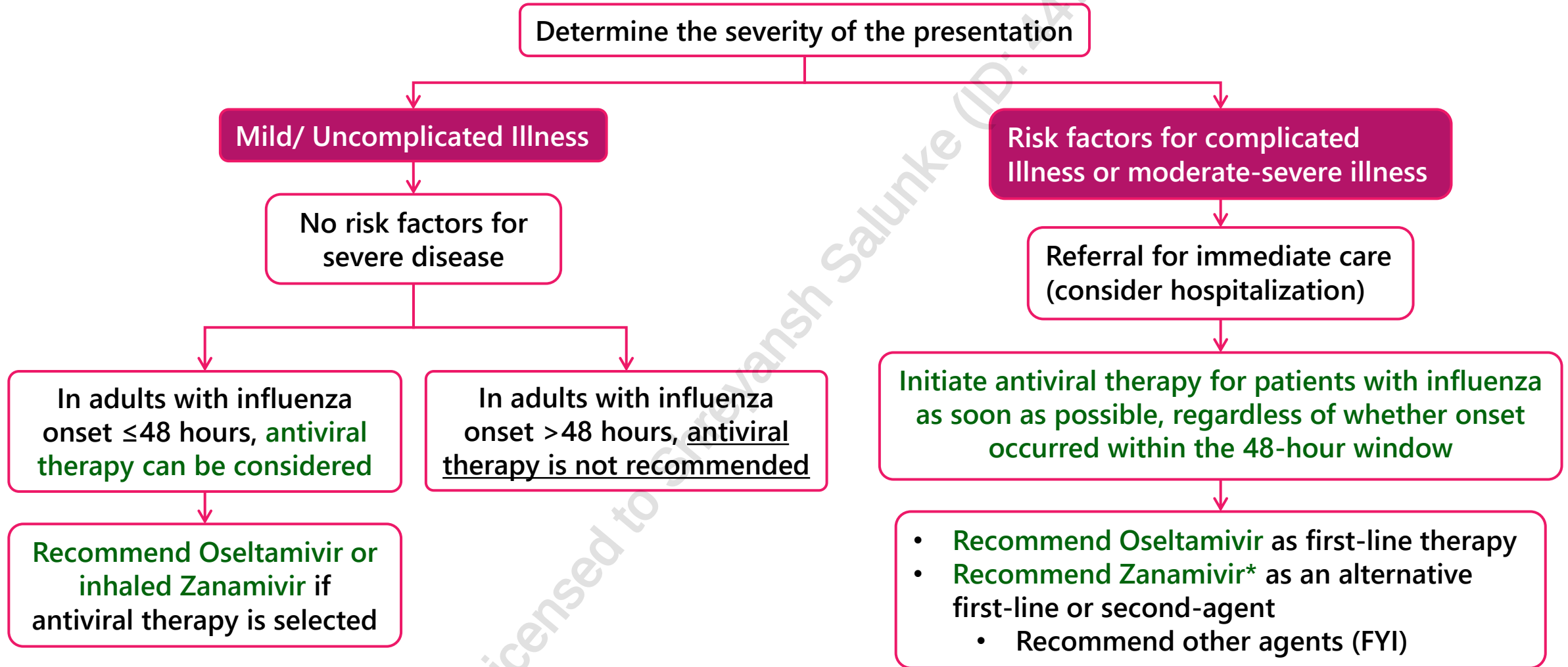
TREATMENT OF INFLUENZA/ SUSPECTED INFLUENZA IN PATIENTS <18 YEARS OLD



**Antiviral treatment may be considered on a case-by-case basis if symptoms are present for > 48 hours.*

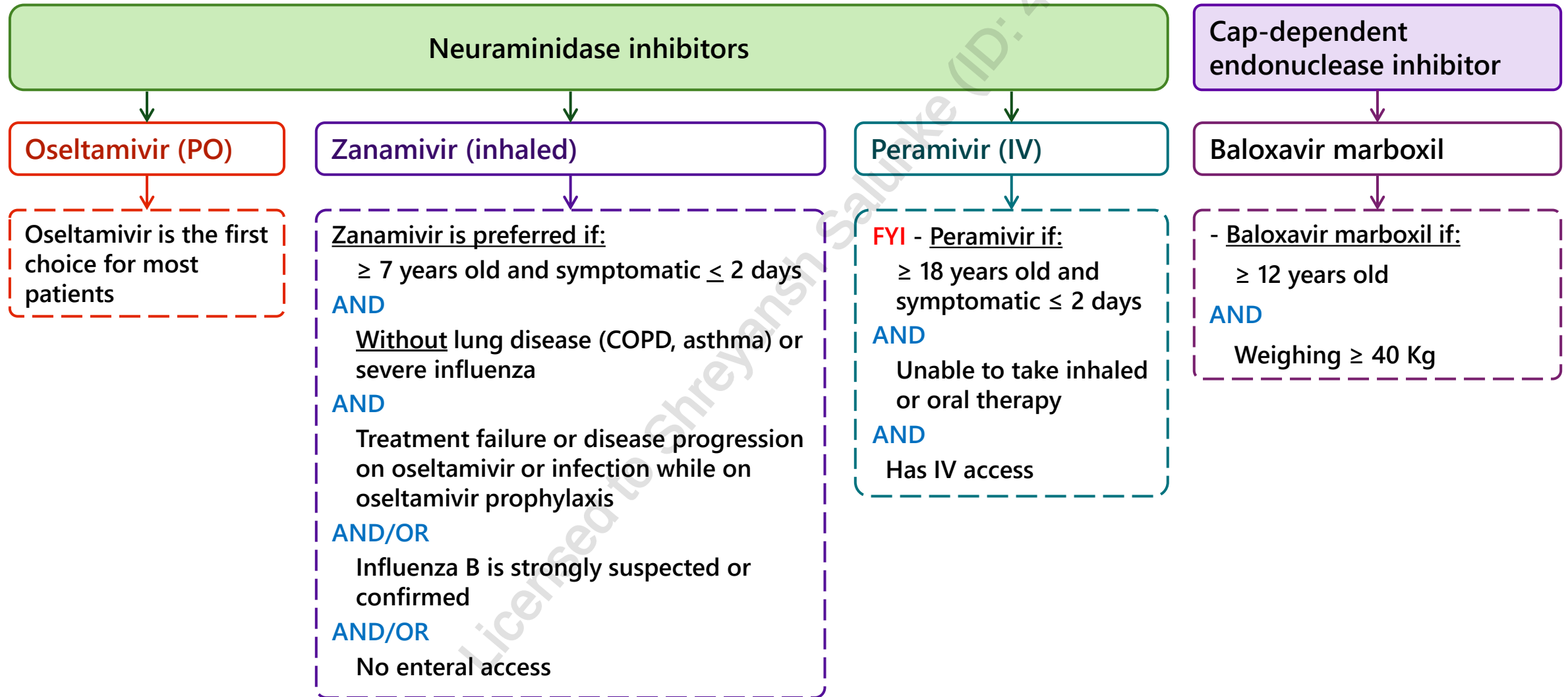
***Zanamivir is recommended in patients without lung disease (COPD, asthma) or severe influenza who have not responded to oseltamivir due to insufficient antiviral effect, in cases where influenza B is highly probable or confirmed.*

TREATMENT OF INFLUENZA/ SUSPECTED INFLUENZA IN PATIENTS ≥18 YEARS OLD



**Zanamivir is recommended in patients without lung disease (COPD, asthma) or severe influenza who have not responded to oseltamivir due to insufficient antiviral effect, in cases where influenza B is highly probable or confirmed.*

PHARMACOLOGIC THERAPY: ANTIVIRAL AGENTS



PHARMACOLOGIC THERAPY: ANTIVIRAL AGENTS

Osetamivir (Tamiflu®)

- **Mechanism of action:** Neuraminidase inhibitor
- **Indication:**
 - **Treatment of influenza** in patients > 1 year old or pregnant (any trimester) with symptoms of less than 48 hours duration (or with risk factors/severe disease)
 - Treatment Dose: 75 mg PO BID x 5 days in adults, 30-75mg PO BID x 5 days in children (weight-based dosing)
 - **Prevention of influenza** A and B in adults and children >1 year who have close contact with an infected individual
 - Chemoprophylaxis Dose: 75 mg PO daily x 10 days in adults, 30-75mg PO daily x 10 days in children (weight-based) (Dose is FYI)
- **Renal dose adjustment is required**
- **Side effects:** nausea/vomiting, neuropsychiatric effects, headache
- **Drug-drug/drug-food interactions:**
 - Avoid administration with LAIV – Given at least 2 weeks before or 48 hours after antiviral treatment

PHARMACOLOGIC THERAPY: ANTIVIRAL AGENTS

Zanamivir (Relenza®)

- **Mechanism of action:** Neuraminidase inhibitor
- **Indication:**
 - **Treatment of uncomplicated influenza** A and B in patients ≥ 7 years old with symptoms of ≤ 48 hours without lung disease or severe illness, and when oseltamivir treatment has been ineffective, and/or strong suspicion of influenza B
 - Treatment dose in adults and children ≥ 7 years: 10 mg (two 5 mg inhalations) PO BID x 5 days
 - **Prevention influenza** in patients ≥ 7 years old
 - Chemoprophylaxis dose in adults and children ≥ 7 years: 10 mg (two 5 mg) PO once daily x 10 days
- **No dose adjustment in renal impairment**
- **Contraindications:** allergy to milk protein/lactose
- **Precaution/warning:** Do not use in the treatment or prophylaxis of patients with underlying airway disease
- **Side effects:** bronchospasm, headache, dizziness, cough
- **Drug-drug/drug-food interactions:**
 - Avoid administration with LAIV -> Given at least 2 weeks before or 48 hours after antiviral treatment

PHARMACOLOGIC THERAPY: ANTIVIRAL AGENTS

FYI

Peramivir (Rapivab®)	Baloxavir marboxil (Xofluza®)
• Mechanism of action: Neuraminidase inhibitor	• Mechanism of action: Cap-dependent endonuclease inhibitor leading to viral genome transcription inhibition
• Indication: Treatment of acute uncomplicated influenza in patients ≥ 18 years old with symptoms of ≤ 48 hours duration	• Indication: treatment of uncomplicated influenza in patients ≥ 12 years old with symptoms of ≤ 48 hours duration who are healthy OR at high risk of developing influenza complications
• Treatment Dose: 600 mg IV x 1 dose • Renal dose adjustment in CrCl < 50 mL/min • Administer after dialysis with dose based on renal function	• Treatment Dose: • 40kg to < 80 kg: 40mg PO once as single dose • ≥ 80 kg: 80mg PO once as single dose
• Side effects: anaphylaxis, hives, SJS, diarrhea, neuropsychiatric events	• Side effects: anaphylaxis, hives, urticaria
• Drug–drug/drug-food interactions: • Avoid administration with LAIV -> Given at least 2 weeks before or 48 hours after antiviral treatment	• Drug–drug/drug-food interactions: • Avoid co-administration with polyvalent cations-containing laxatives, calcium-fortified foods (like dairy), antacids or oral supplements of mono/divalent cations • Substrate of P-gp and UGT 1A3, but limited risk for drug interactions • Concurrent use with LAIV has not been determined

PHARMACOLOGIC THERAPY: PROPHYLAXIS

NAIs antiviral prophylaxis is efficacious and well-tolerated for:

1. **Seasonal: two weeks of NAI prophylaxis given at the time of inactivated influenza vaccine (known as bridging prophylaxis)**
 - Prevention in high-risk individuals during the first two weeks following vaccination
 - Prevention for people with severe immune deficiencies or others who might not respond to influenza vaccination, such as persons receiving immunosuppressive medications
 - Prevention for people at high risk for complications from influenza who cannot receive influenza vaccine due to a contraindication
2. **Post-exposure prophylaxis: Antiviral administration within 48 hours of contact with confirmed or suspected influenza**
 - Early treatment is better due to the developing resistance of oseltamivir due to PEP
3. **Outbreak: Chemoprophylaxis + antiviral treatment in closed facilities**
 - Prevention of influenza among residents of institutions, such as long-term care facilities, during influenza outbreaks in the institution

INFLUENZA VACCINE

NACI Statement:

- Influenza activity typically peaks during December but can start as early as October or as late as February, lasting 12 – 20 weeks
- It is recommended to get the injection in the fall to give the most protection for the winter

Government of Canada Recommendations:

- Encourages all healthy persons aged 6 months and older without contraindication to be offered a vaccine
- Influenza vaccination is recommended for those at high risk of severe disease, healthcare workers, essential service providers, and individuals in contact with avian influenza-infected poultry
- Patients 9 years and older should receive 1 dose annually
- Patients 6 months to < 9 years who are unvaccinated should receive 2 doses 28 days apart during their first season, then 1 dose annually

FLU VACCINES AND EGG ALLERGY

- The government of Canada's National Advisory Committee on Immunizations has recommended the administration of a full dose of any influenza vaccine to patients with egg allergy without prior skin testing in any setting
- However, immunizers administering the vaccine should be prepared for and have the necessary equipment to respond to a vaccine emergency

FLU VACCINES AND GUILLAIN-BARRÉ SYNDROME (GBS)

- Although the evidence for a causal relationship between influenza vaccination and GBS is inadequate, NACI recommends the following:
 - Individuals who have **developed GBS within 6 weeks of a previous flu vaccine should not receive another flu vaccine** unless another cause was discovered for GBS
- Must weight the risk of a recurrent episode of GBS with the benefits of flu vaccination

INFLUENZA VACCINE CHOICE

FYI

Age	Choice	Comment
6 – 23 months	IIV-Adj, IIV-SD, IIV-cc	No longer any preference between quadrivalent and trivalent in children (as per NACI updates 2024-2025)
2 – 17 years	IIV-SD, IIV-cc, LAIV	Avoid LAIV in severe/uncontrolled asthma, pregnancy, immunocompromise, ASA use (risk of Reye's syndrome)
18 – 59 years	IIV-SD, IIV-cc, RIV, LAIV	Avoid LAIV in healthcare workers, pregnancy, patients who are immunocompromised, and those with high-risk chronic conditions (ex. pulmonary, cardiac, renal, etc.) LAIV may not be as effective in adults as other options
60 – 64 years	IIV-SD, IIV-cc, RIV	
≥65 years	IIV-Adj, IIV-SD, IIV-HD, IIV-cc, RIV	HD preferred over SD

IIV3-Adj: adjuvanted trivalent inactivated influenza vaccine

IIV3-SD: standard-dose trivalent inactivated influenza vaccine

IIV4-cc: quadrivalent mammalian cell culture-based inactivated influenza vaccine IIV4-HD: high-dose quadrivalent inactivated influenza vaccine

IIV4-SD: standard-dose quadrivalent inactivated influenza vaccine

RIV4: quadrivalent recombinant influenza vaccine

LAIV4: quadrivalent live attenuated influenza vaccine

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Influenza symptoms e.g. fever (should not last > 3 days), cough, arthralgia, myalgia, exhaustion/weakness (residual cough may last for weeks)	Improve/resolve	4-7 days
Influenza complications e.g. pneumonia	Prevent	Throughout therapy

Safety Endpoints		
Parameter	Degree of Change	Timeframe
ADRs e.g. nausea, dizziness, headache	Minimize/eliminate	Duration of therapy

TAKEAWAYS

- Influenza is transmitted from an infected individual via droplets to a non-infected host
 - Transmission can be restricted via non-pharmacological measures of hand hygiene, wearing a mask, staying home when sick, etc
- Infection symptoms can range from asymptomatic, mild, moderate, severe and death
- Influenza testing is not recommended for individuals without risk factors for complications. However, patients who are at high risk or exhibit red flag symptoms should be referred without delay

Treatment:

- Patients <18 with moderate-severe symptoms or complicated illness should be treated with Oseltamivir or Zanamivir
- Patients > 18 with risk factors for complicated illness or moderate-severe illness should be referred immediately to care and treated with Oseltamivir or Zanamivir
- Zanamivir is preferred if: ≥ 7 years old and symptomatic < 2 days AND without lung disease (COPD, asthma) or severe influenza AND treatment failure or disease progression on oseltamivir or infection while on oseltamivir prophylaxis AND/OR influenza B is strongly suspected of confirmed AND/OR No enteral access

TREATMENT TAKEAWAY

NAIs	Oseltamivir	Zanamivir
Dosage Form:	Tablets PO	Inhaled PO
Treatment Regimen:	BID for 5 days in adults and children ≥ 1 years old	PO BID x 5 days In adults and children ≥ 7 years old
Chemoprophylaxis Regimen:	Once daily for 10 days in adults and children	
Dose Adjustments:	Dose adjustments requirement in renal impairment	No renal dose adjustment required
Warnings:	Neuropsychiatric side effects	CI in allergy to milk protein, and underlying airway disease
Vaccines:	Avoid administration with LAIV -> Given at least 2 weeks before or 48 hours after antiviral treatment	

CASE 1

AS is a 69-year-old female presenting to the pharmacy with dyspnea, myalgia, and rhinorrhea. She reports her symptoms started yesterday and have worsened today. Her 5-year-old grandson is also experiencing rhinorrhea, but his symptoms are not as severe. She has not checked for fever but reports feeling chills. She reports receiving the high dose quadrivalent influenza vaccine (IIV4-HD) last week. Her blood glucose from this morning was 8.2 mmol/L. AS has a past medical history of type 2 diabetes mellitus, hypertension, chronic kidney disease (last CrCl = 45mL/min), and chronic obstructive pulmonary disease. She takes metformin 500 mg po BID, linagliptin 5 mg po daily, insulin glargine 8 units qhs, ramipril 5 mg po daily, Advair® (fluticasone propionate/salmeterol) 250 mcg/50mcg one puff BID, and vitamin D 1000 units po daily. She weighs 63 kg and has an allergy to sulfa drugs. She lives in a retirement home.

1. What agent should be recommended to treat AS' myalgia?
 - a) ASA
 - b) Acetaminophen
 - c) Ibuprofen
 - d) Naproxen



CASE 1

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2. Which of the following is the main driver for suggesting antiviral therapy for AS?

- a) Age $\geq$ 65
- b) Female sex
- c) Hypertension
- d) Institutional setting



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3. What is the best option for antiviral treatment of AS' influenza?

- a) Baloxavir marboxil 80 mg PO x 1 dose
- b) Oseltamivir 30 mg PO BID x 10 days
- c) Zanamivir 5 mg 2 inhalations BID x 5 days
- d) Oseltamivir 30 mg PO BID x 5 days



CASE 1

AS is a 69-year-old female presenting to the pharmacy with dyspnea, myalgia, and rhinorrhea. She reports her symptoms started yesterday and have worsened today. Her 5-year-old grandson is also experiencing rhinorrhea, but his symptoms are not as severe. She has not checked for fever but reports feeling chills. She reports receiving the high dose quadrivalent influenza vaccine (IIV4-HD) last week. Her blood glucose from this morning was 8.2 mmol/L. AS has a past medical history of type 2 diabetes mellitus, hypertension, chronic kidney disease (last CrCl = 45mL/min), and chronic obstructive pulmonary disease. She takes metformin 500 mg po BID, linagliptin 5 mg po daily, insulin glargine 8 units qhs, ramipril 5 mg po daily, Advair® (fluticasone propionate/salmeterol) 250 mcg/50mcg one puff BID, and vitamin D 1000 units po daily. She weighs 63 kg and has an allergy to sulfa drugs. She lives in a retirement home.

4. Which of the following BEST describes the rationale for providing antiviral therapy to AS?
- a) Reduced risk of mortality due to myocardial infarction
 - b) Reduced duration of influenza virus shedding
 - c) Reduced duration of influenza-related cough and fever
 - d) Reduced risk of progression to severe disease



CASE 1

AS is a 69-year-old female presenting to the pharmacy with dyspnea, myalgia, and rhinorrhea. She reports her symptoms started yesterday and have worsened today. Her 5-year-old grandson is also experiencing rhinorrhea, but his symptoms are not as severe. She has not checked for fever but reports feeling chills. She reports receiving the high dose quadrivalent influenza vaccine (IIV4-HD) last week. Her blood glucose from this morning was 8.2 mmol/L. AS has a past medical history of type 2 diabetes mellitus, hypertension, chronic kidney disease (last CrCl = 45mL/min), and chronic obstructive pulmonary disease. She takes metformin 500 mg po BID, linagliptin 5 mg po daily, insulin glargine 8 units qhs, ramipril 5 mg po daily, Advair® (fluticasone propionate/salmeterol) 250 mcg/50mcg one puff BID, and vitamin D 1000 units po daily. She weighs 63 kg and has an allergy to sulfa drugs. She lives in a retirement home.

5. In addition to the need for antivirals, which of the following should prompt a referral to another healthcare provider for this patient?

- a) Hyperglycemia
- b) Age
- c) Comorbidity
- d) Dyspnea



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CHANGE LOG

May 2025

- Slide 30: Changed q3 dosing for oseltamivir in renal impairment

June 2025

- Slide 14: Added note on case-by-case indication for antivirals if symptoms presents > 48 hours

July 2025

- Slides 14 and 26: Edited the note on age

September 2025

- Slide 18: Updated guidance on zanamivir use in treatment of uncomplicated influenza to align with slide 16

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 5: Added FYI
- Slide 15: Minor re-phrasing updates
- Slide 25: Updated/re-formatted monitoring parameters